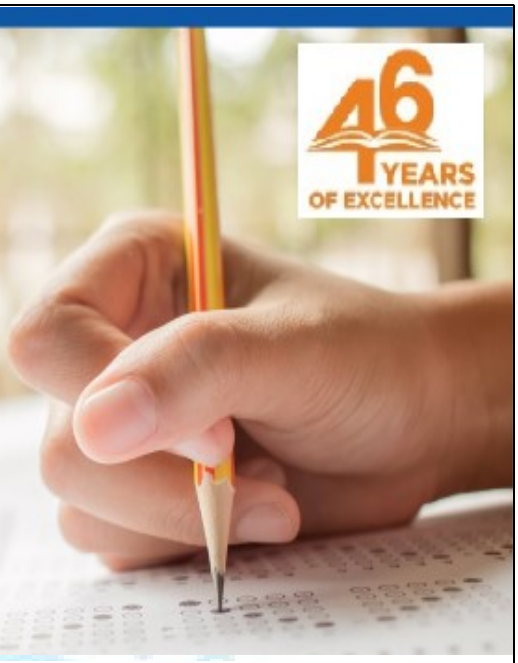
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 16-04-26
Time: 3 Hrs Max. Marks: 180

16-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-6(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 21)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 22 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 25 – 30)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 38)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 39 – 41)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 42 – 47)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 48 –51)	Questions with **Comprehension Type** With **Numerical value type** (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
Total				**17**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. If $L=\text{Lt}_{n\to\infty}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{x^4e^{ix}\left(e^{ix}\cos x-i\sin 2x\right)}{\ln\left(\sum_{r=0}^{n}\frac{x^2}{r!}\right)}\cdot dx\left(i=\sqrt{-1}\right)$ then the value of $2L$ equals

A) π^2-8 B) $\pi-8$ C) π^2-4 D) $\frac{\pi^2}{2}-2$

2. Let $A(Z_1),B(Z_2),C(Z_3)$ be the vertices of ΔABC such that $|Z_1-Z_2|=3,Z_1^2=Z_2Z_3,Z_2^2=Z_1\cdot Z_3,(Z_1Z_2Z_3\neq 0)$ then the area ΔABC is equal to

A) $\frac{3\sqrt{3}}{4}$ B) $\frac{9\sqrt{3}}{4}$ C) $3\sqrt{3}$ D) $\frac{\sqrt{3}}{4}$

3. The value of the sum $\sum_{n=1}^{\infty}\cot^{-1}\left(\frac{\left(n^2+2n\right)\left(n^2+2n+1\right)+1}{2n+2}\right)$ is equal to $\sec^{-1}\left(\frac{\sqrt{5}}{\lambda}\right)$ then the value of λ equals

A) 1 B) 2 C) $\frac{1}{2}$ D) $\sqrt{2}$

4. L_1,L_2 are two parallel lines C_1,C_2,C_3 are three circles. C_1 is tangent to L_1 and L_2. Now C_2 touches C_1 externally and tangent to L_1 also C_3 touches C_1 externally and tangent to L_2. Both C_2,C_3 lie between the lines L_1 and L_2. If the radii of C_2 and C_3 are $5\sqrt{2}$ and $10\sqrt{2}$ respectively and touch externally, then the diameter of C_1 is equal to

A) $15\sqrt{2}$ B) 15 C) 20 D) 40

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SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. Let a line parallel to the axis of the parabola $y^2 = 4x$ be drawn through the point $P(\alpha - 5, \alpha)(\alpha \neq 0)$ to meet the parabola at Q. Equation of tangent at Q is $x - 2y + 4 = 0$. Let T be a point on the tangent. Let M, N be feet of perpendiculars on SQ and the directrix respectively from point T. (is focus)

Which is/are correct ?

A) The value of α is equal to -4

B) If PA, PB are tangents to the parabola touching at A and B then the distance of orthocenter of ΔPAB from origin is $\sqrt{17}$

C) If PA, PB are tangents to the parabola touching if at A and B then are of ΔPAB is $20\sqrt{5}$

D) If $QM = 3$ then TN can be 2

6. If $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{(\sin^3 x - \cos^3 x - \cos^2 x)(\sin x + \cos x + \cos^2 x)^{2025}}{(\sin x)^{2027}(\cos x)^{2027}} dx$ is equal to $\frac{(a+\sqrt{b})^{2026} - (1+\sqrt{c})^{2026}}{d+1} (a, b, c, d \in N)$ then which is/are correct

A) $a + b = 5$
B) $c = 6$
C) d is odd
D) Number of possible integral divisors of d is 15

7. Let $A\left(\frac{1}{2}, 1, 3\right)$ and $C\left(3, \frac{17}{4}, \frac{13}{2}\right)$ be the vertices of ΔABC and equation of angle bisector of $\angle ABC$ is $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{2}$, then which is/are correct?

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A) equation of side BC is $\frac{x-1}{4}=\frac{2y-5}{7}=\frac{z-2}{9}$

B) equation of side BC is $\frac{x-1}{2}=\frac{y+2}{7}=\frac{z+1}{3}$

C) co-ordinates of vertex B is (α,β,γ) then $\alpha+\beta+\gamma=15$

D) co-ordinates of vertex B is (α,β,γ) then $\alpha+\beta+\gamma=22$

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. Let

$(a+1)(b+1)(c+1)(d+1)=3$

$(a+2)(b+2)(c+2)(d+2)=7$

$(a+3)(b+3)(c+3)(d+3)=13$

$(a+4)(b+4)(c+4)(d+4)=21$

Then the number of positive integral divisors of $(a+5)(a+6)(b+5)(b+6)(c+5)(c+6)(d+5)(d+6)$ is equal to ____

9. Let $f(x)$ be a function defined by $f(x)=\begin{cases}|x^5-4x^3+2|\cos(\pi x), x\geq 0\\ x^3-\sin x-|(x^2+8x+12)x^2|, x<0\end{cases}$ and if the number of points of discontinuities is p and the number of points of non differentiability is q then the value of p^2+pq+q^2 equals

10. Let $PQRS$ be a tetrahedran. Given that $PQ=6, PR=4, PS=3, QR=4, QS=5$ and $RS=2$. If A and AB respectively mid points of edges PQ and RS and if the value of AB equals K, then $10K^2$ ____

11. In a recent examination conducted in a teacher's class,
i) 5 students earned a score of atleast 95%
ii) 13 students earned a score of atleast 90%
iii) 27 students earned a score of atleast 85%
iv) 50students earned a score of atleast 80%

If the number of students earned a score of atleast 80% and less than 90% is m and if the number of students earned a score atleast 80% and less than 85% is n then the value of $m+n$ equals

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12. Pawan Kalyan wrote down units digit of each of the first 2025 positive squares i.e., $1,4,9,6,5,6,\ldots$. If the arithmetic mean of these numbers is p and variance of these numbers is q then the value of $(p+\sqrt{q})^2+(p-\sqrt{q})^2$ equals K so that $[K]$ equals ____ . ([.] is GIF)

13. Let $f(N)=\sum_{n=2}^{N}\frac{\log_2\left(1+\frac{1}{n}\right)}{\left(\log_2^n\right)\left(\log_2^{(n+1)}\right)}$ then the integral part of reciprocal of $f(31)\cdot f(63)\cdot f(127)\cdot f(255)\ldots\ldots 10$ terms is equal to ____ .

SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 14 and 15:

Let Z_1,Z_2,Z_3 be complex numbers such that $Z_1^2+Z_2^2+Z_3^2=Z_1Z_2+Z_2Z_3+Z_3Z_1$ and $|Z_1+Z_2+Z_3|=21$. Given that $|Z_1-Z_2|=2\sqrt{3},|Z_1|=3\sqrt{3}$ then answer the following

14. The value of $|Z_2|^2+|Z_3|^2$ is equal to ____

15. The value of $Z_1\vec{Z}_2+Z_2\vec{Z}_1+Z_2\vec{Z}_3+Z_3\vec{Z}_2+Z_1\vec{Z}_3+Z_3\vec{Z}_1$ is equal to ____

Paragraph-II for Question Nos. 16 and 17:

Let $f(x),f'(x),f''(x)$ are continuous on $[0,\ell n2]$ and $f(0)=0,f'(0)=3,f'(\ell n2)=4,f(\ell n2)=6$ and $\int_0^{\ln 2}e^{-2x}f(x)dx=3$. Also $g(x)$ be a continuous function satisfying $\int_0^1 g(x)\left(4x^2-g(x)\right)dx=\frac{4}{5}$ now answer the following

16. The value of $\int_0^{\ln 2}e^{-2x}f''(x)dx$ equals ____ .

17. The value of $m+n$ if the arithmetic mean of $g\left(\frac{1}{2}\right),g\left(\frac{1}{2^2}\right),\ldots,g\left(\frac{1}{2^{10}}\right)$ is $\frac{1}{m}\left(1-\frac{1}{2^n}\right)(m,n\in N)$ is equal to ____ .

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PHYSICS **MAX.MARKS: 60**

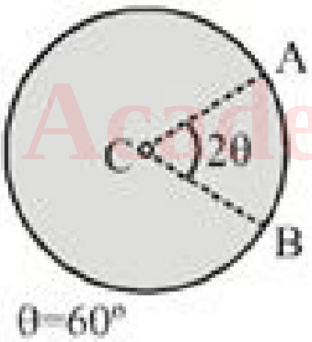
SECTION–I (Maximum Marks: 12)

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Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

18. What should be the approximate kinetic energy of an electron so that its de-Broglie wavelength is equal to the wavelength of X-ray of maximum energy, produced in an X-ray tube operating at 24800 V ? (given that $h = 6.6\times10^{-34}$ joule - sec, mass of electro $n = 9.1\times10^{-31}$ kg)

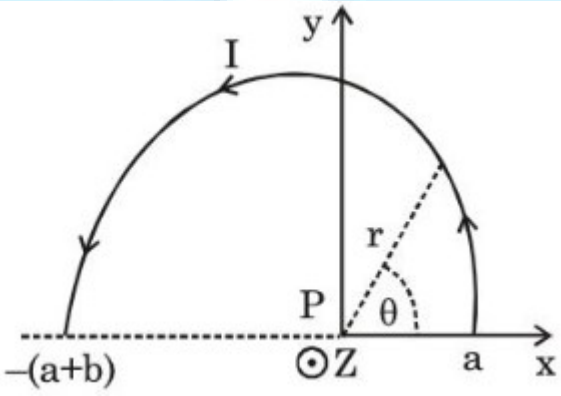
A) 600 Ev B) 36.5 eV C) 6000 eV D) 300 eV

19. A rocket is launched from point A on earth. It reaches point B on earth with its velocity antiparallel to its initial velocity. The angle subtended by line AB at the center is 2θ. The satellite moving close to surface of earth has time period of revolution T_0. Radius of earth is R. Approximate ratio of time taken by the rocket from A to B to the period of revolution (if complete orbital motion would have been possible) of satellite is



A) 1/3 B) 2/3 C) 1/2 D) 3/4

20. A wire segment is bent into the shape of an Archimedes spiral (see figure).



The equation that describes the curve in the range $0 \le \theta \le \pi$

$r(\theta) = a + \frac{b}{\pi}\theta$, for $0 \le \theta \le \pi$

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Where θ is the angle from x -axis in radians. Point P is at the origin. I is the current. Magnetic field at point P is

A) $\frac{\mu_0 Ib}{2a}$ B) $\frac{\mu_0 I}{2\pi}\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}$ C) $\frac{\mu_0 I}{4}\left(\frac{1}{a}-\frac{1}{b}\right)$ D) $\frac{\mu_0 I}{4}\ln\left(1+\frac{a}{b}\right)$

21. A solid cone is place on horizontal surface has height h, R and apex angle θ as shown. If gravitational potential energy of the cone does not change as position of the cone is changed from figure (A) to figure (B), then

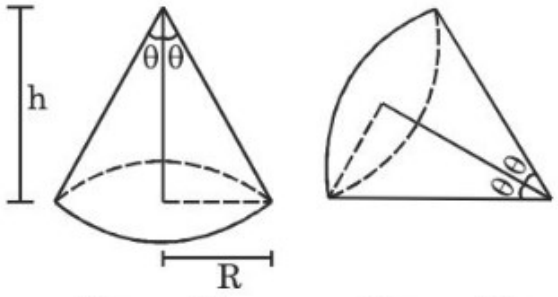


Figure (A) Figure (B)

A) $\theta = \sin^{-1}\left(\frac{2}{3}\right)$ B) $\theta = \frac{\pi}{4}$ C) $\frac{h}{R}=1$ D) $\frac{h}{R}=2\sqrt{2}$

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

22. A sound wave of frequency f_0 travels horizontally to the right. It is reflected from a large vertical plane surface moving to left with a speed v. The speed of sound in medium is C.

A) The wavelength of the incident wave on the plane changes to $\frac{C^2}{(C+v)f_0}$

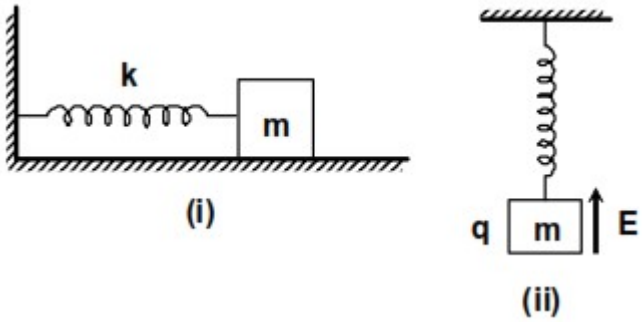
B) The wavelength of the reflected wave is $\frac{C(C-v)}{f_0(C+v)}$

C) The no. of beats heard by an observer at the left of the surface is $\frac{2vf_0}{C-v}$

D) Speed of reflected sound with respect to the plane is $C-v$.

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23. A particle of mass m is executing SHM as shown in diagram (1) and (2). In diagram (2) particle contains charge q such that $Eq = mg$. If their velocities are same at mean position and let A_1 and A_2 be their amplitudes and T_1 and T_2 be their time periods respectively then



A) $A_1 = A_2$ B) $A_1 < A_2$ C) $T_1 = T_2$ D) $T_1 < T_2$

24. A conducting spherical shell of inner radius a and outer radius b is made up of a material of resistivity ρ and negligible dielectric activity. At time $t = 0$, material of shell is electrically neutral and a charge q_0 is some how located at the centre of shell (i.e., at time $= 0$, there are no induced charges, they will appear afterward). Choose correct statement(s)

A) $E = \dfrac{q_0 - Q_t}{4\pi\epsilon_0 r^2}$ where r is distance of point under consideration from centre of shell

B) Current through shell at this moment (discussed in option A) is $I = \dfrac{q_0 - Q}{\epsilon_0 \rho}$

C) At time t we can write $\dfrac{dQ_t}{q_0 - Q_t} = \dfrac{dt}{\epsilon_0 \rho}$

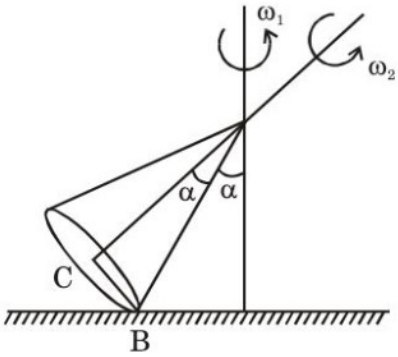
D) At time t, value of Q_t given by $Q_t = q_0\left(1 - e^{\frac{1}{\epsilon_0 \rho}}\right)$

SECTION-III (Maximum Marks : 24)

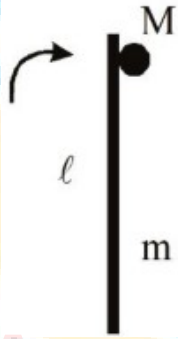
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Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

25. A uniform solid conc (height $h = 4m, \alpha = 30^\circ$) is inclined against a vertical axis as shown in the figure. The cone rotates about its own axis as well as rotates about the vertical axis with angular speeds marked in the diagram. If the cone does not slip at point B ; find $\dfrac{\omega_1}{\omega_2}$.

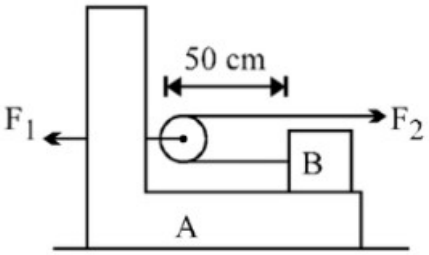
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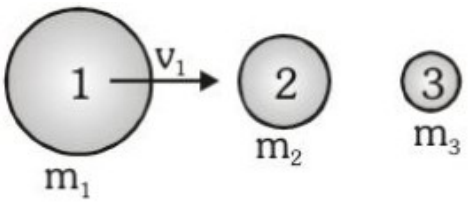
26. A uniform stick of mass 18 kg and length 3 m spins around on a frictionless horizontal plane, with its Centre of mass stationary. A mass M is placed on the plane, and the stick collides elastically with it, as shown (with the contact point being the end of the stick). What should M (in kg) be so that after the collision the stick has translational motion, but no rotational motion?



27. A 1 kg block B rests as shown on a bracket A of same mass. Constant forces $F_1 = 20$ N and $F_2 = 8$ N start to act at time $t = 0$ when the distance of block B from pulley is 50 cm . Time when block B reaches the pulley is $\frac{3}{N}$ sec. Determine 'N'.

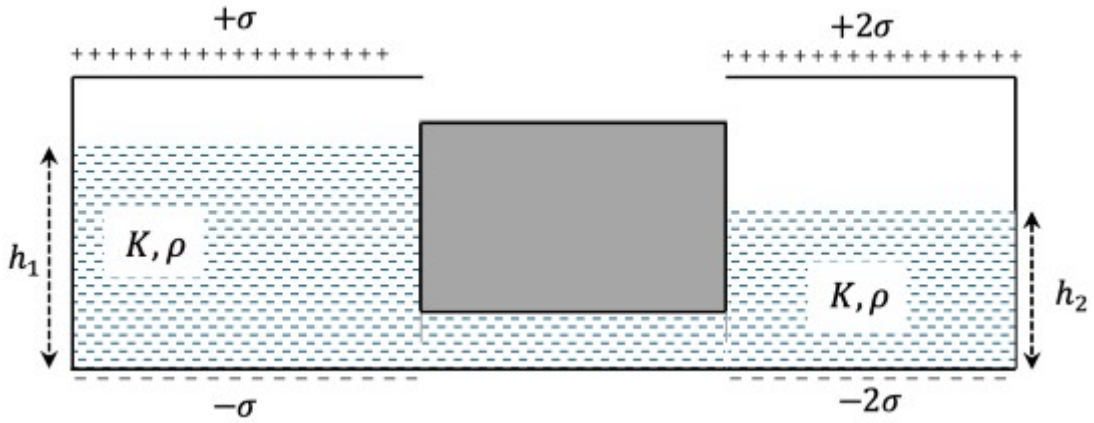


28. The centres of the spheres 1,2 and 3 lie on a single straight line. Sphere 1 is moving with an (initial) velocity v_1 directed along this line and hits sphere 2 .Sphere 2 , acquiring after collision a velocity v_2, hits sphere 3 . Both collisions are absolutely elastic. What must be the mass of sphere 2 (in kg) for the sphere 3 to acquire maximum velocity (the masses m_1 and m_3 of spheres 1 and 3 are 9 kg & 1 kg respectively) ?

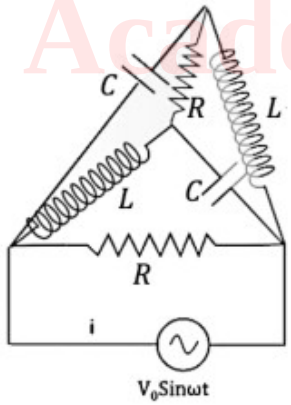


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29. A liquid of density ρ, Dielectric constant $K = 2$, is filled in two communicating containers. The base and a top plates are uniformly charged to charge densities σ and 2σ of opposite nature, as shown in the figure. The height difference of liquid levels $|h_1 - h_2|$ is $\frac{n}{8} \times \frac{\sigma^2}{\epsilon_0 \rho g}$. Determine the value of n.



30. The sides of a tetrahedron contain two capacitors of capacitance C each, two inductors of inductance L each and two resistors of R each, as shown in the figure. Given, $\omega = \frac{1}{\sqrt{LC}}, R = \sqrt{\frac{L}{C}}$. Determine the maximum current through the AC source in ampere.
($R = 50\Omega, V_0 = 100$ V)



SECTION - IV (Maximum Marks : 12)

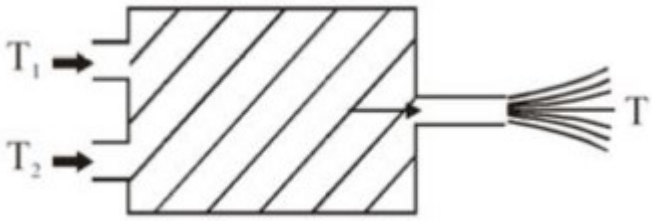
This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 31 and 32:

A self-contained machine only inputs two equal steady streams of hot and cold water at temperature T_1 and T_2. Its only output is a single high-speed jet of water. The heat capacity per unit mass of water, C , may be assumed to be independent of temperature.

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The machine is in a steady state and the kinetic energy in the incoming steams is negligible.



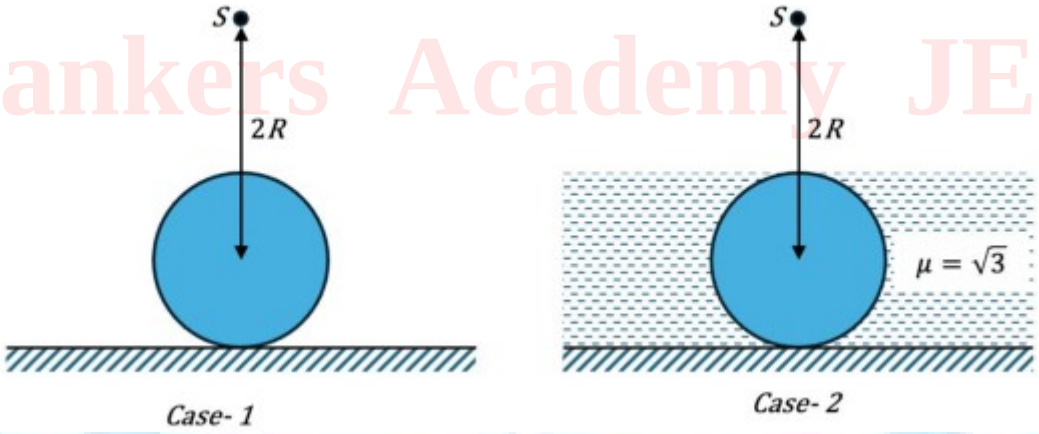
$(T_1 = 400\ \text{K}, T_2 = 300\ \text{K}, C = 4\ \text{J/K})$

31. What is the speed of the output jet of the liquid (in m/s) if $T = 348.5\ \text{K}$

32. What is the maximum possible speed of the output jet of the liquid (in m/s)

Paragraph-II for Question Nos. 33 and 34:

An opaque sphere of radius R lies on a horizontal plane. At a height 2 R vertically above the centre of the sphere there is a point source S (Case-1). In Case-2, a transparent liquid of refractive index $\sqrt{3}$ is filled above the planesuch that the sphere is just covered with the liquid.



33. The area of the shadow on the plane is $n \times 4\pi R^2$ in Case-1. Determine n.

34. The area of the shadow on the plane is $n \times 4\pi R^2$ in Case-2. Determine n.

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CHEMISTRY **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

35. Natural polymer among the following is
A) Vulcanized rubber B) Rayon
C) Cellulose acetate D) Silk

36. A metal exists in two allotropic forms, α and β, having a face centred cubic and hexagonal closest packed lattice respectively. Identity the correct statement.
A) Coordination number is higher for the α-form
B) Density is higher for the β - form
C) Distance between closest octahedral voids is same for both forms
D) Unit cell volume is same for both forms.

37. $M^{n+}(aq) + KI \rightarrow \underset{ppt}{X}\downarrow \xrightarrow{\text{Excess } KI}$ ppt. remains insoluble in excess KI solution.
A) $Pb^{2+}(aq)$ B) $Cu^{2+}(aq)$ C) $Bi^{3+}(aq)$ D) $Hg^{2+}(aq)$

38. Which of the following compounds are ionic
a) SnF_4 b) $SnCl_4$ c) PbF_2
d) CF_4 e) PbF_4
A) a, b and d B) a, c and d C) a, c and e D) a, b, c and e

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. Among the following correct statement is/are
A) Morphine narcotics are obtained from opium pappy
B) Narcotic analgesics in medicinal dose can be used for the relief of post operative pains, cardiac pains.
C) Morphine $\xrightarrow{\text{excess}(CH_3CO)_2O/H^+}$ Codine
D) Morphine $\xrightarrow[II)CH_3I]{I)\ NaOH\ 1eq}$ Heroin

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40. Which of the following are correct for a chemical reaction?

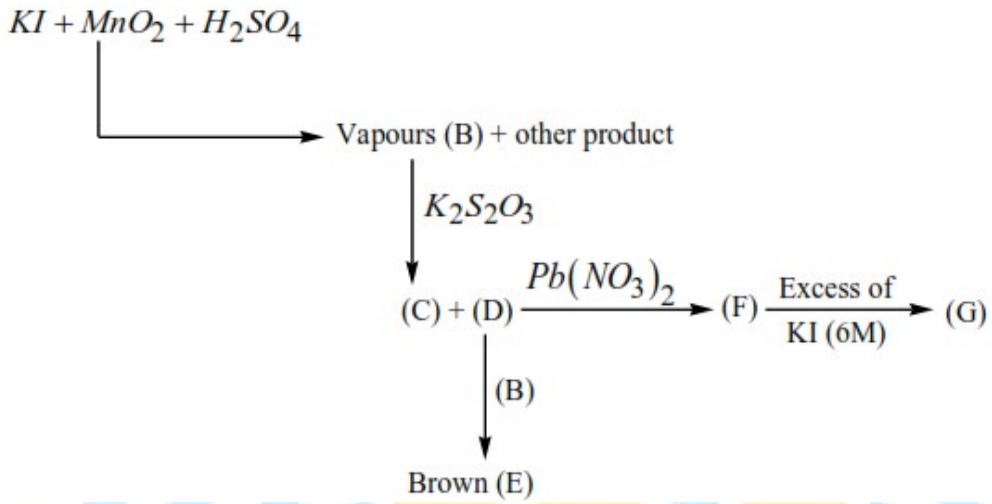
A) When a catalyst is introduced, at constant temperature and the order of reaction does not change, both rate of reaction and rate constant increases.

B) On increasing temperature of an elementary reaction, both rate of reaction and rate constant increases.

C) On changing volume, at constant temperature the rate of reaction will not be affected for a zero order reaction.

D) The temperature dependence of rate constant is higher in reactions with higher activation energies.

41.



Select the incorrect statement(s)

A) Vapours of 'B' combine with excess of chlorine gives periodic acid in the presence of water

B) The hybridization of ' G ' is sp^3

C) 'B' in solid state, it's dissociaiton gives cation and anion and the sum of hybrid orbitals in cation and anion equals to 10

D) 'B' & 'E' oxidises hypo gives same products

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

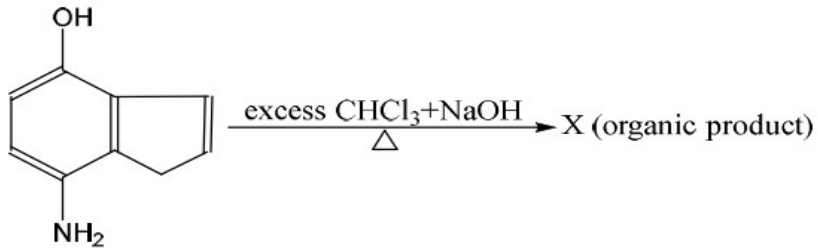
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

42.



In this reaction sufficient high temperature maintained. Gram molecular weight (in g) of X is ____

(G.A.W.of H is1, C is12, N is14, O is16, Na is 23, Cl is 36)

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43.

$$\underset{(\text{saturated acyclic compound})}{X} \xrightarrow{V_2O_5,500^\circ C} \text{Meisitylene} \xleftarrow{\text{Red hot } Fe \text{ tubes}} 'Y' \text{gas}$$

$$X \xrightarrow[\text{Monochlorination}]{+Cl_2/h\vartheta} 'Z'(\text{Monochloro isomers})$$

Gram molecular weight (in g) of Y is ' M '.
Product of ' Z ' and ' M ' is ____ .

44. An ideal solution of 2 mol chlorobenzene and 3 mol bromobenzene is enclosed in a cylinder - piston arrangement, and the piston is pulled outwards very slowly allowing equilibrium to be maintained at all times. The vapour pressure of pure chlorobenzene and pure bromobenzene are 300 torr and 200 torr respectively. If the vapour phase mole fraction of chlorobenzene is Y_1 in the first traces of vapours formed, and Y_2 when the last traces of liquid is about to be vapourized, what is $\frac{4Y_1}{Y_2}$? (Assume the process to be isothermal).

45. How many of the following statements are correct for the reaction

$$CrO_5 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + H_2O + O_2$$

(i) 2 moles CrO_5 liberates 3.5 moles of O_2.
(ii) 1 mole CrO_5 liberates 3.5 moles of O_2.
(iii) All ' O ' in CrO_5 have an oxidation state of -1 .
(iv) Cr in CrO_5 is reduced.
(v) Cr in CrO_5 is oxidized.
(vi) Some ' O ' in CrO_5 is reduced.
(vii) Some ' O ' in CrO_5 is oxidized.
(viii) Some ' O ' in CrO_5 is neither reduced nor oxidized.
(ix) 3 moles of H_2O are formed per mol of CrO_5 reacted.
(x) Cr has an oxidation state of +5 in CrO_5.

46. In $\left(Cu(NH_3)_4\right)^{2+}$ and $(Ni(CN)_4)^{4-}$ complex ions. The sum of axial 'd' electrons are present in both the complex ions equal to

47. The number of molecule(s) or ion(s) from the following having non-planar structure is

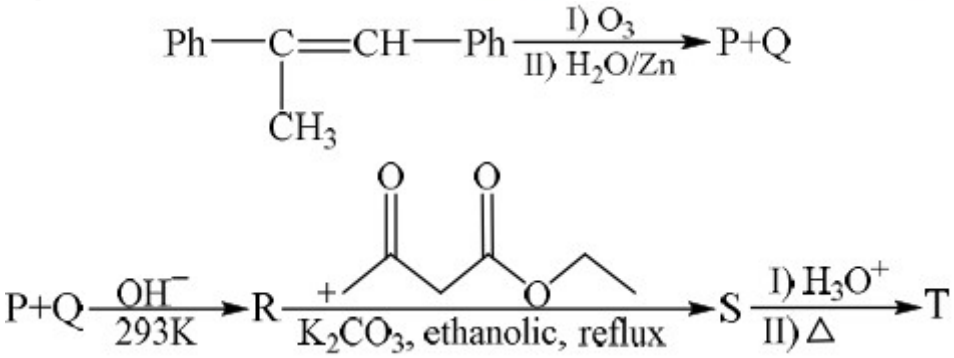
$$NO_3^-, H_2O_2, BF_3, PCl_3, XeF_4, SF_4, XeO_3, PH_4^+, SO_3, [Al(OH)_4]^-, ClF_3$$

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SECTION - IV (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Paragraph-I for Question Nos. 48 and 49:



48. Sum of degree of unsaturation of R, S and T is ____ .

49. Gram molecular weight (in g) of T is ____ .

Paragraph-II for Question Nos. 50 and 51:

Rusting of iron involves the following electrode processes : $Fe(s) \rightarrow Fe^{2+}_{(aq)} + 2e^-$ and $O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-_{(aq)}$

The standard electrode potentials are : $E^{2+/Fe}_{Fe^\circ} = -0.44$ V, $E_{O_2/OH^-} = 0.40$ V

Take $\frac{RT\ln 10}{F} = 0.06, \log 2 = 0.31, \log 3 = 0.48$

Consider temperature to be $25^\circ C$, with $K_w = 10^{-14} M^2$. Density of water, and aqueous solution $= 1 gmL^{-1}$. Atmospheric pressure is 1 bar, and the atmosphere is $79\% N_2, 20\% O_2$ and 1% other gases by volume. K_C for $SO_2(aq) + H_2O(l) \rightleftharpoons H^+_{(aq)} + HSO_3^-(aq)$ is 1.6×10^{-2} and for $HSO_3^-(aq) \rightleftharpoons H^+_{(aq)} + SO_3^{2-}(aq)$ is 10^{-8}

50. Calculate the electrode potential (in mV) for rusting, (as described above) if the water is pH neutral and $[Fe^{2+}]$ is $0.01M$.

51. What would be the electrode potential (in mV) measured for rusting (as described above) in an industrial region where concentration of $Fe^{2+}_{(aq)}$ was 0.01 M and the concentration of $SO_2(aq)$ in the water was found to be 0.1 M at equilibrium. Other atmospheric condition remain unchanged. (Assume SO_2 is the only species responsible for acidity of water).

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